Design of high fidelity combinatorial libraries

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Abstract

Recent tools for de novo design have revolutionized the way we propose novel proteins and the types of challenges we are able to tackle. While these tools have dramatically increased the hit rate of functional designs, zero-shot generations often require further optimization to enhance the desired properties. We present a novel method for generating high fidelity diverse combinatorial libraries. Traditional library generation methods rely on error prone PCR or natural fragment shuffling to generate diversity. Such approaches do not take into account the filtering of designs through structure prediction models or other desired oracles. We demonstrate the ability to post-train inverse folding models to design fragments with filter pass rates of 53.5% and libraries which can easily scale beyond 10^{12} individually addressable variants. Our post-training framework enables control over the search space by centering the library around a reference sequence with tunable mutation distance. Once designed, we demonstrate how individual variants are assembled with simple molecular biology techniques (golden gate assembly). Our cell-free assay platform enables us to screen enzymes at the scale of 10^3 measurements per day, each with replicates. The libraries generated are easily much larger than most traditional and modern high throughput assays are able to screen, so we demonstrate how the library can be sub sampled smartly to optimize enzymes. For a de novo designed PETase we demonstrate our platform finds a variant with 250x increase in catalytic activity.

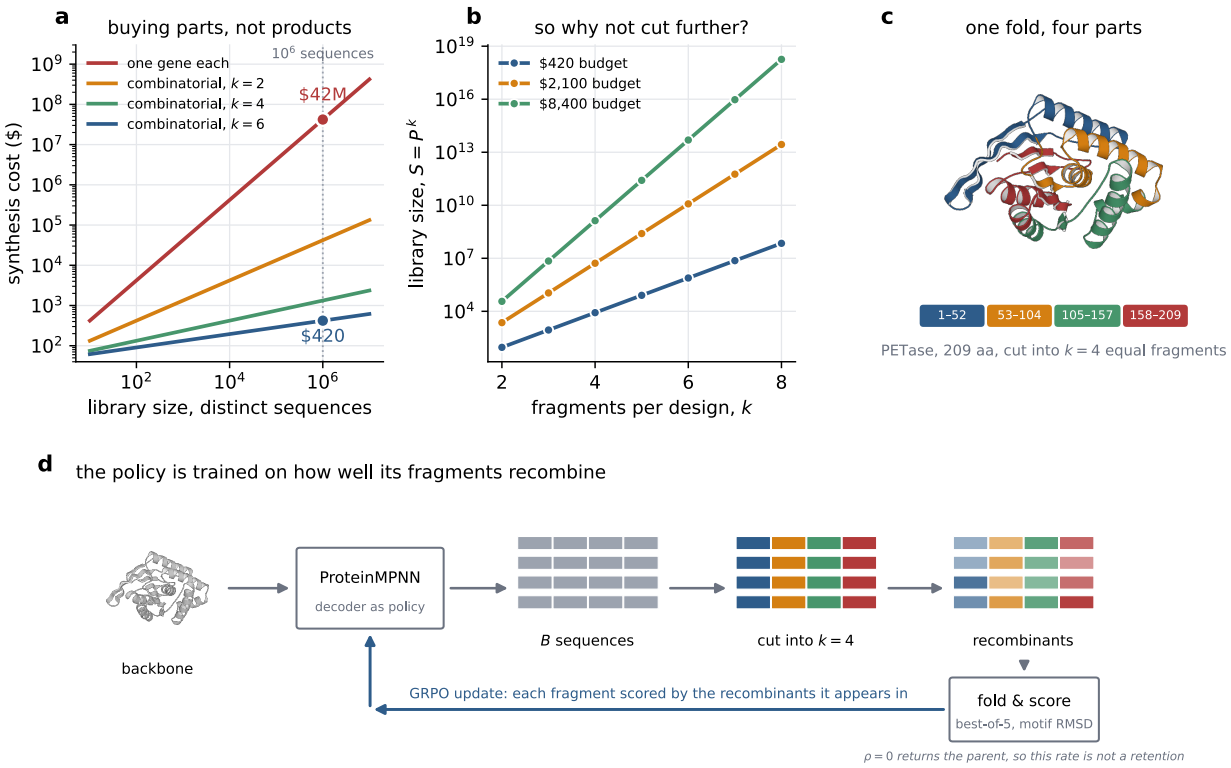


Figure 1: **A combinatorial library, and the policy that makes one assemblable.** **(a)** Ordering N distinct sequences as individual genes costs $N \times \$42$ at eBlock pricing for a 200-aa enzyme; ordering the same N as a combinatorial library costs $N^{1/k} \times \$42$, because what is bought is k sets of parts rather than N products. At 10^6 sequences that is \$42M against \$420. **(b)** At a fixed synthesis budget the reachable library size grows as $S = P^k$, which invites cutting into ever more fragments. **(c)** The object: a 209-residue PETase cut into $k = 4$ equal fragments, coloured by slot. Equal lengths are chosen so that assembly is robust, not because the boundaries are structurally meaningful. **(d)** The method. A backbone conditions the ProteinMPNN decoder, which is treated as a policy; the sequences it emits are cut at the fragment boundaries, recombined across parents, and folded best-of-5. Each fragment is scored by the recombinants it appears in, and that per-fragment reward drives the GRPO update. The policy is therefore optimised for how well its fragments *recombine*, not for how good any single sequence is.